

Project Report on

Retrieval-Augmented Generation (RAG) System using FastAPI & Gemini

Project Objective:

An LLM-based solution to assist users in drafting project proposals from project documents.

The aim of this project is to build a Retrieval-Augmented Generation (RAG) system that extracts fine details from IFB/IFPQ PDF documents and allows users to query this information via a chatbot-like interface. The system uses:

- FastAPI for the backend API
- Google Gemini API for answering user questions based on extracted content
- PyMuPDF for PDF parsing

Project Structure:

```
.
├── app/
│   ├── __init__.py
│   ├── gemini.py          # Gemini API integration
│   └── utils.py           # PDF text extraction
├── data/
│   └── IFB_document.pdf   # Source documents (PDFs)
├── sample_questions.txt   # List of questions
├── fine_prints.txt        # Generated summary of key fine prints
├── chat_response.txt      # Output answers to the questions
├── main.py                # FastAPI entry point
├── requirements.txt
└── README.md
```

Project Flow:

1. PDF Parsing

On hitting /fine-prints, the system loads all PDFs from the data/ folder and extracts their textual content using PyMuPDF.

2. Summarization

The extracted content is passed to the Gemini API, which summarizes or extracts key fine prints. The output is saved to fine_prints.txt.

3. Question Answering (Chat)

When /chat is called:

- It loads fine_prints.txt as context.
- It reads questions from sample_questions.txt.
- It uses Gemini to generate answers for each question.
- The results are saved in chat_response.txt.



FastAPI Endpoints:



GET /

Description:

Health check route.



GET /fine-prints

Description:

Triggers text extraction and fine print summarization from PDFs.

Flow:

- Extract text from all PDF files in /data.
- Use Gemini to extract or summarize fine prints.
- Save result to fine_prints.txt.

```
127.0.0.1:8000/fine-prints
127.0.0.1:8000/fine-prints
Pretty-print
{
  "fine_prints": "Okay, here's a breakdown of the most important rules, deadlines, and restrictions extracted from the IFB and IFPQ documents, focusing on what a legal assistant would need to flag:\n\n**IFB # 03A3974 (Hot Mix Asphalt Concrete Repair)**\n\n  **Purpose:** Hot Mix Asphalt (HMA) concrete grinder dig-out repair and replacement paving in Glenn, Butte, Sutter, Yuba, Colusa, Placer, Nevada, El Dorado, and Sierra Counties.\n\n  **Submission Deadline:** April 24, 2025, 2:00 p.m. Pacific Time.\n\n  **Small Business (SB) Participation Goal:** Caltrans is committed to meeting the State's twenty-five percent (25%) Small Business (SB) participation goal. SB's, MB's, SB-PW's are encouraged to submit bids.\n\n  **DVB Participation Required:** 3% participation required.\n\n  **Prevailing Wages:** Required if the total bid amount exceeds $15,000.\n\n  **Contact Person:** Esmeralda Rojo, esmeralda.rojo@dot.ca.gov, (279) 234-2457. All questions must be submitted in writing.\n\n  **DIR Registration:** Bidder must have a valid and current registration with the Department of Industrial Relations (DIR) at the time of bid submittal and for the duration of the Agreement.\n\n  **Contractor's License:** A - General Engineering Contractor or C-12 Earthwork and Paving Contractor and C-31 Construction Zone Traffic Control Contractor license issued by the CSLB.\n\n  **Bidder's Bond (ADM-2010):** Required for bids $461,000 and greater. Bidder shall submit a bid bond (executed by an admitted surety insurer) with bid. The bidder's bid bond shall be no less than 10 percent (10%) of bidder's total bid.\n\n  **Payment Bond Requirement:** The successful bidder will be required to provide, prior to commencement of work under a Service Request, a Payment Bond for one hundred percent (100%) of the total Service Request, if the Service Request price is $25,000 or more.\n\n  **Insurance:** Certificate of Insurance is required within 10 working days of notification of award. Specific insurance requirements and coverage limits are detailed in Exhibit E of the Proposed Form of Agreement.\n\n  **California Civil Rights Laws Certification (DOT ADM -0076):** Required if the contract is $100,000 or more.\n\n  **Darfur Contracting Act Certification (DOT ADM-0077):** Required.\n\n  **Iran Contracting Act Certification (DOT ADM-0078):** Required.\n\n  **Required Documents Checklist:** See Attachment 3.\n\n  **Bid Labeling:** Specific mailing package/envelope labeling instructions in Section C.21.B. Packages not properly labeled may be rejected.\n\n  **No alternate agreement language accepted.**\n\n  **IFPQ # 01A6494 (Window, Insulated Panel, Rain Gutter, and Painting Services):**\n\n    **Purpose:** Window, Insulated Panel, Rain Gutter, and Down Drain Replacement/Installation Services and Painting Services in Mendocino and Lake County.\n\n    **Submission Deadline:** April 24, 2025, 2:00 PM Pacific Time.\n\n    **Estimated Contract Value:** Award of contracts for the acquisition of construction, including the erection, construction, alteration, repair, or improvement of any State structure, building, road, or other State improvement of any kind, that has an estimated value of greater than $5,000.00, but not more than $461,000.00.\n\n    **Small Business (SB) Set-Aside:** ONLY certified Small Businesses (SB), including Micro Businesses (MB), Small Business for the purpose of Public Works (SB-PW), or Disabled Veteran Business Enterprise (DVB) registered with the California Department of General Services (DGS) are eligible to bid.\n\n    **Mandatory Site Inspection:** April 17, 2025, 10:00 AM at Lakeport HS, 3218 Hill Road East, Lakeport, CA 95453. Attendance is mandatory.\n\n    **Prevailing Wages:** Required if the total bid amount exceeds $15,000.\n\n    **Contact Person:** Wendy Kwong, Wendy.Kwong@dot.ca.gov, (279) 789-4474. All questions must be submitted in writing.\n\n    **Contractor's License:** Class B - General Building Contractor license issued by the CSLB.\n\n    **DIR Registration:** Required.\n\n    **Subcontractors:** May only subcontract to qualified DGS Certified SB/MB/SB-PW or DVB.\n\n    **Payment Bond:** If the successful bidder's bid exceeds $25,000, the successful bidder shall furnish a Payment Bond for not less than 100 percent (100%) of the total amount payable by the terms of this Agreement.\n\n    **Insurance:** Certificate of Insurance is required within 10 working days of notification of award. Specific insurance requirements and coverage limits are detailed in Exhibit E of the Proposed Form of Agreement.\n\n    **California Civil Rights Laws Certification (DOT ADM -0076):** Required if the contract is $100,000 or more.\n\n    **Darfur Contracting Act Certification (DOT ADM-0077):** Required.\n\n    **Iran Contracting Act Certification (DOT ADM-0078):** Required.\n\n    **Required Documents Checklist:** See Attachment 3.\n\n    **Submission Method:** Email to Wendy.Kwong@dot.ca.gov, subject line must include IFPQ number, title, and \"PRICE QUOTATION SUBMITTAL\".\n\n  }\n}"
```

fine_prints.txt: ( File Output)

```
File Edit Selection View Go ... data
readme.md fine_prints.txt
fine_prints.txt
1 Okay, here's a breakdown of the most important rules, deadlines, and restrictions extracted from the IFB and IFPQ docu
2
3 **IFB # 03A3974 (Hot Mix Asphalt Concrete Repair)**
4
5 * **Purpose:** Hot Mix Asphalt (HMA) concrete grinder dig-out repair and replacement paving in Glenn, Butte, Sutter,
6 * **Submission Deadline:** April 24, 2025, 2:00 p.m. Pacific Time.
7 * **Small Business (SB) Participation Goal:** Caltrans is committed to meeting the State's twenty-five percent (25%)
8 * **DVB Participation Required:** 3% participation required.
9 * **Prevailing Wages:** Required if the total bid amount exceeds $15,000.
10 * **Contact Person:** Esmeralda Rojo, esmeralda.rojo@dot.ca.gov, (279) 234-2457. All questions must be submitted in
11 * **DIR Registration:** Bidder must have a valid and current registration with the Department of Industrial Relation
12 * **Contractor's License:** A - General Engineering Contractor or C-12 Earthwork and Paving Contractor and C-31 Cons
13 * **Bidder's Bond (ADM-2010):** Required for bids $461,000 and greater. Bidder shall submit a bid bond (executed by
14 * **Payment Bond Requirement:** The successful bidder will be required to provide, prior to commencement of work und
15 * **Insurance:** Certificate of Insurance is required within 10 working days of notification of award. Specific insu
16 * **California Civil Rights Laws Certification (DOT ADM -0076):** Required if the contract is $100,000 or more.
17 * **Darfur Contracting Act Certification (DOT ADM-0077):** Required.
18 * **Iran Contracting Act Certification (DOT ADM-0078):** Required.
19 * **Required Documents Checklist:** See Attachment 3.
20 * **Bid Labeling:** Specific mailing package/envelope labeling instructions in Section C.21.B. Packages not properly
21 * **No alternate agreement language accepted.**
```

✓ POST /chat

Description:

Loads fine_prints.txt and sample_questions.txt, and answers each question using Gemini.

Expected sample_questions.txt format:

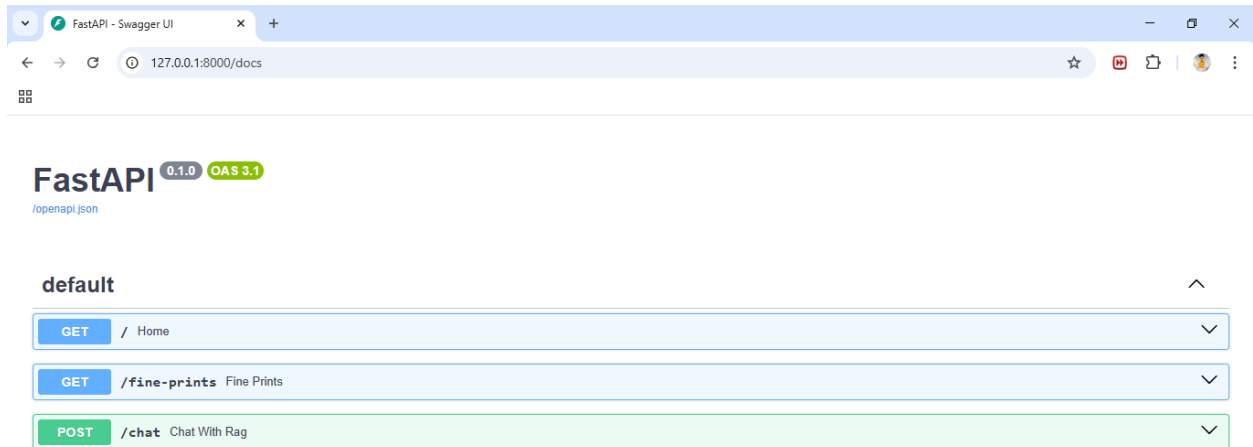
What is the deadline for bid submission?

Who is the contact person for the IFPQ?

What licenses are required to submit a bid for the IFB?

Is small business certification required for the IFPQ?

- Visit /docs (Swagger UI) to test /chat
- Open:
 - `http://127.0.0.1:8000/docs`
 - Then:
 -
 - Find the /chat endpoint (method: POST)
 -
 - Click "Try it out"



FastAPI - Swagger UI

127.0.0.1:8000/docs#/default/chat_with_rag_chat_post

GET /fine-prints Fine Prints

POST /chat Chat With Rag

Parameters

No parameters

Try it out

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value

Schema

"string"

FastAPI - Swagger UI

127.0.0.1:8000/docs#/default/chat_with_rag_chat_post

200

Response body

```
{
  "chat": [
    {
      "question": "What is the deadline for bid submission?",
      "answer": "Both IFB # 03A3974 and IFPQ # 01A6494 have a submission deadline of **April 24, 2025, 2:00 PM Pacific Time.**"
    },
    {
      "question": "Who is the contact person for the IFPQ?",
      "answer": "Wendy Kwong, Wendy.Kwong@dot.ca.gov, (279) 789-4474."
    },
    {
      "question": "What licenses are required to submit a bid for the IFB?",
      "answer": "To submit a bid for IFB # 03A3974 (Hot Mix Asphalt Concrete Repair), you need one of the following licenses issued by the CSLB:\n\nA - General Engineering Contractor\nC-12 Earthwork and Paving Contractor\nC-31 Construction Zone Traffic Control Contractor"
    },
    {
      "question": "Is small business certification required for the IFPQ?",
      "answer": "Yes, small business certification is required for the IFPQ # 01A6494. ONLY certified Small Businesses (SB), including Micro Businesses (MB), Small Business for the purpose of Public Works (SB-PW), or Disabled Veteran Business Enterprise (DVB-E) registered with the California Department of General Services (DGS) are eligible to bid."
    }
  ]
}
```

Response headers

```
content-length: 1069
content-type: application/json
date: Tue, 13 May 2025 05:07:27 GMT
server: uvicorn
```

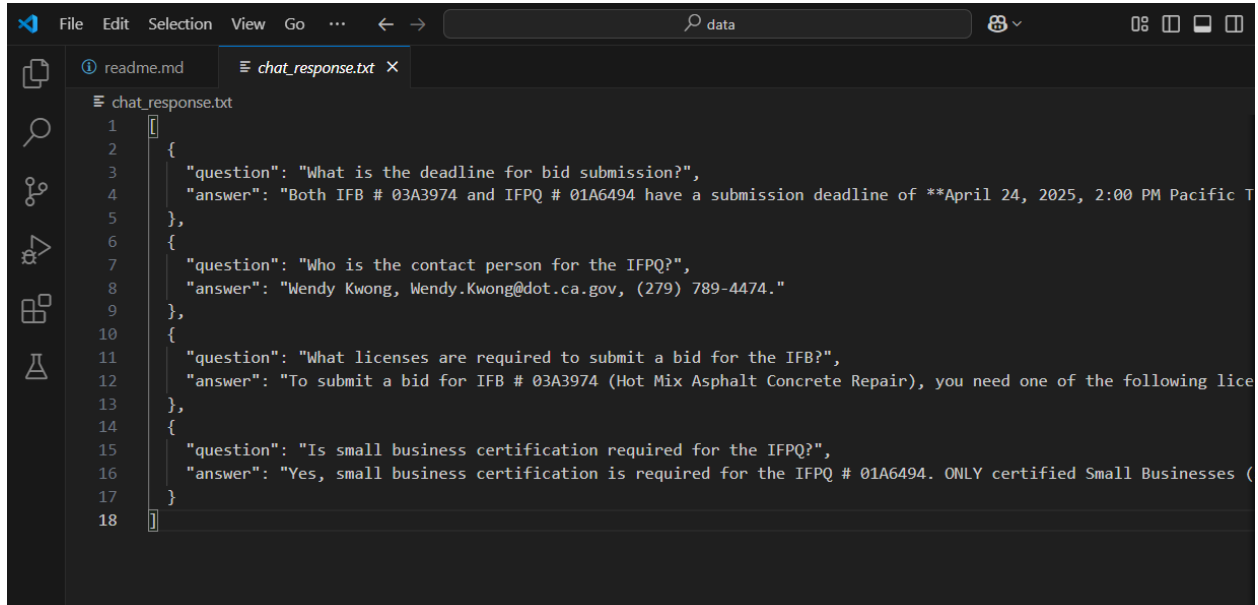
Responses

Code	Description	Links
200	Successful Response	No links

Media type

Chat_response.txt:

 File Output



```
1 {
2   {
3     "question": "What is the deadline for bid submission?",
4     "answer": "Both IFB # 03A3974 and IFPQ # 01A6494 have a submission deadline of **April 24, 2025, 2:00 PM Pacific T
5   },
6   {
7     "question": "Who is the contact person for the IFPQ?",
8     "answer": "Wendy Kwong, Wendy.Kwong@dot.ca.gov, (279) 789-4474."
9   },
10  {
11    "question": "What licenses are required to submit a bid for the IFB?",
12    "answer": "To submit a bid for IFB # 03A3974 (Hot Mix Asphalt Concrete Repair), you need one of the following lice
13  },
14  {
15    "question": "Is small business certification required for the IFPQ?",
16    "answer": "Yes, small business certification is required for the IFPQ # 01A6494. ONLY certified Small Businesses (
17  }
18 }
```

Technologies & Libraries Used

Library	Purpose
FastAPI	Web API framework to serve endpoints like /fine-prints and /chat
Uvicorn	ASGI server to run the FastAPI app
PyMuPDF (fitz)	For extracting text content from PDF documents
google-generativeai	Gemini API integration for summarization and Q&A
Pydantic	Input validation (used for request schemas)
python-dotenv	To load your Gemini API key from a .env file
JSON module	For writing and reading structured question-answer pairs

Files and Their Role

File	Description
<code>main.py</code>	The entry point for the FastAPI server with routes for summarization and chat
<code>app/utils.py</code>	Contains logic to extract text from PDF files
<code>app/gemini.py</code>	Contains functions to interact with the Gemini API
<code>sample_questions.txt</code>	User questions (1 per line)
<code>fine_prints.txt</code>	Gemini-generated summary of the IFB/IFPQ document
<code>chat_response.txt</code>	Stores the final questions and answers
<code>requirements.txt</code>	All necessary libraries for the project
<code>.env</code>	Holds the Gemini API key (not committed to GitHub)
<code>data/</code>	Folder where PDF documents are stored